Problem 1 /16

TAKEHOME. Submit at the start of the final exam at 8:30 am on Monday, December 8. Use R for all quantitative calculations, including code and output in all cases. Follow all standards and best-practices used throughout the semester.

This problem refers to the `onions` data set, available on Moodle. This set includes information about mean yield per pound and planting densities of three varieties of onions.

(a) Is the `Variey` variable quantitative or qualitative? What is the level of measurement? Briefly explain your answers.

(b) Compute the five-number summary and inter-quartile range of planting densities for onions in this set.

(c) Determine the 60^{th} percentile of onion yields in this set.

(d) Compute a level 95% confidence interval for the mean yield *without* using the `t.test` function. Briefly explain your choice of method (t or z). Identify the point estimate, margin or error, and endpoints of the interval.

Problem 2 /20

Use R for all quantitative calculations. Include code and output in all cases.

This problem refers to the `cheese_sample` data set, available on Moodle. This set gives lactic acid concentrations (units unknown) and taste ratings for 7 cheeses.

Lactic acid	0.86	1.15	1.30	1.49	1.63	1.68	1.72
Taste	12.3	14.0	25.9	6.4	18.0	34.9	26.5

(a) In just a sentence or two, explain the circumstances under which it would be appropriate to calculate the correlation coefficient of these two variables. No R code is needed for this part.

(b) Assume the conditions in part (b) are met and determine the correlation of these two variables.

(c) Find the equation of the least-squares regression line. Use `Lactic` as the explanatory variable.

(d) What is the predicted taste rating of a cheese with a lactic acid concentration of 1.30? If this calculation isn't appropriate, briefly explain why.

(e) What is the residual of the cheese with lactic acid concentration of 1.30? Briefly interpret this number in ordinary human language.